

1. Understanding Vector Embeddings

- Vector embeddings convert text, images or other data into numerical vectors that preserve semantic meaning.
- Similar concepts are represented by vectors located close together in high-dimensional space.
- Embeddings enable semantic search, recommendation systems, similarity matching and retrieval-augmented generation (RAG).
- Unlike keyword search, embeddings understand context and meaning.
- Typical workflow: Raw text → Embedding model → Dense vector → Vector database → Similarity search.

2. Google Colab Setup

- Install required libraries: google-generativeai, faiss-cpu, sentence-transformers, streamlit and pyngrok.
- Import Google Gemini SDK, FAISS, NumPy and SentenceTransformer.
- SentenceTransformer creates embeddings while FAISS performs efficient similarity search.

3. RAG and Gemini Workflow

- User submits a query.
- Relevant documents are retrieved from a knowledge base using vector search.
- Retrieved context is supplied to Gemini.
- Gemini generates grounded, accurate and context-aware answers.
- RAG reduces hallucinations by providing external knowledge.

4. Gemini-Powered AI Applications

- Gemini supports text, image, code and multimodal inputs.
- Maintains conversation context for natural interactions.
- Can be integrated with Streamlit to build interactive AI applications.
- Works seamlessly with RAG systems for domain-specific knowledge.

5. Introduction to FAISS

- FAISS (Facebook AI Similarity Search) is a library for fast vector similarity search.
- Supports millions of embeddings using efficient indexing.
- Provides nearest-neighbor search for semantic retrieval.
- Common use cases include chatbots, search engines and recommendation systems.

6. FAISS Architecture

- Data is converted into embeddings.
- Embeddings are stored inside a FAISS index.
- Incoming queries are embedded and compared against stored vectors.
- Most similar vectors are returned and passed to an LLM for response generation.

7. What is Streamlit?

- Streamlit is a Python framework for building web applications with minimal frontend code.
- Supports real-time interaction and rapid deployment.
- Ideal for AI dashboards, demos and machine learning applications.

8. Why Use Streamlit?

- Very little boilerplate code.
- Quick prototyping of AI products.
- Interactive dashboards and data visualization.
- Easy integration with machine learning models and APIs.

9. Building AI Interfaces with Streamlit

- Accept user input through text boxes and widgets.
- Upload files for document analysis.
- Display AI-generated responses and visualizations.
- Acts as the frontend while Gemini and FAISS handle intelligence.

10. How Embeddings Work

- Input text is processed by an embedding model.
- The model converts text into a numerical vector.
- Similarity search compares vectors using cosine similarity or Euclidean distance.
- Closest vectors correspond to semantically similar information.
- Advantages include faster retrieval, better context matching and improved search relevance.

End of Notes